ALGEBRA 1 · RELATIONS AND FUNCTIONS

Function Detector

Explore & Reflect

Name _____ Date _____ Period _____

Open **Function Detector** on Mr. K's Classroom: Algebra 1 → Relations and Functions → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Do you think a circle is a function? Why or why not?

2. A relation contains (1, 4) and (1, 7). Is it a function?

2 Explore use the tool and record what you see

Sweep each relation

Drag the vertical line all the way across. Domain and range appear after a full sweep.

Relation	Passes the vertical line test?	Function?	Domain	Range
Line				
Parabola				
Sideways parabola				
Circle				
Absolute value				
Vertical line				

My points

Plot points by clicking the grid. Watch the mapping diagram.

	Your points	What the mapping diagram shows
A function		
Not a function		

3 Write the rule fill in each blank from what you observed

1. A relation is a function when each _____ has exactly one _____.

2. If any vertical line hits a graph in more than one point, the graph _____ a function.

3. In a mapping diagram of a function, no input has _____.

4. The domain is the set of all _____. The range is the set of all _____.

4 Check no tool: show your work

1. Is {(2, 3), (4, 3), (6, 1)} a function?

2. Is {(−1, 0), (2, 5), (−1, 4)} a function?

3. Give the domain and range of $y = x^2 + 1$.

4. Is $x = y^2$ a function?

5 Reflect

Why does the vertical line test work?